Montclair State University

System Analysis & Design Project Report ~

Social Service

Project Name: ***FairFlow***

Zakary Macejka

System Analysis & Design CSIT317_02FA25

Professor Khan

11/30/2025

Executive Summary

The Idea~

FairFlow is a data driven demand forecasting and food distribution optimization app for local community pantries.

The Problem~

Food banks face an ongoing challenge in matching fluctuating food donations with community demand across different neighborhoods. Because demand varies weekly and depends on socioeconomic conditions, unemployment rates, and seasonal factors, many local pantries experience either food shortages or excess perishable inventory leading to spoilage. Currently, most food allocation decisions are manual and reactive, relying on prior experience rather than data analytics. This leads to inefficiencies, increased waste, and inequitable service coverage.

The Solution~

Predict neighborhood level food demand using historical distribution, census, and economic data. Next, recommend optimal allocation plans to minimize shortages and spoilage. Use the data to provide interactive dashboards for local pantry organizers to visualize demand hotspots and delivery routes.

Introduction and Background

Food insecurity continues to be one of the most urgent social issues in the United States, impacting more than 47 million people including about 13 million children. This means that 1 in 5 kids in America are affected, according to *Feeding America*. The *U.S. Department of Agriculture (USDA)* further reported that over 17 million American households faced food insecurity in 2023, marking a sharp rise compared to previous years. Although strong food donation networks and charitable organizations such as *Feeding America* and *Second Harvest Heartland* exist, persistent inefficiencies in collection, storage, and distribution prevent food from being delivered equitably to those who need it most.

Second Harvest Heartland has shown how data driven strategies can strengthen hunger relief efforts. By using advanced mapping and logistics tools, the organization has improved its ability to pinpoint regions with the greatest need and streamline food distribution across Minnesota. Likewise, *Feeding America's Map the Meal Gap* study reveals disparities in food access at the county and district level, demonstrating how geographic, economic, and demographic factors shape hunger nationwide. Together, these findings highlight the growing importance of technology based solutions that can make food donation and distribution more efficient, transparent, and fair.

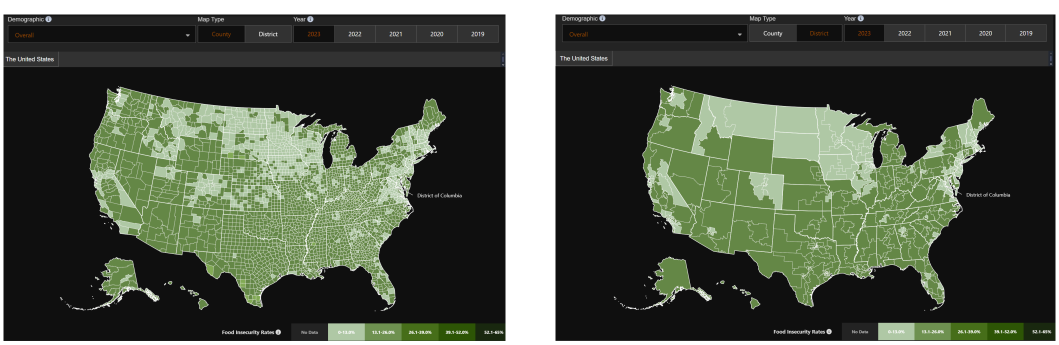


Figure 1. Map of food insecurity rates by U.S. county (2023). Adapted from "Map the Meal Gap," Feeding America, 2023. Retrieved from https://map.feedingamerica.org/county/2023/overall

Introduction and Background

To help address these challenges, FairFlow is proposed as an intelligent, data driven system designed to support food banks, donors, and community organizations in improving the collection and redistribution of surplus food. Leveraging predictive analytics and optimization algorithms, FairFlow seeks to forecast demand, align donations with real time needs, and reduce food waste throughout the supply chain. Functioning as a centralized digital platform, it will enable collaboration among donors, food banks, volunteers, and recipients to ensure that resources reach the communities most in need.

Ultimately, FairFlow's mission is to strengthen the efficiency and equity of food distribution networks by harnessing modern information systems to combat hunger. This project proposal outlines the background, objectives, stakeholder requirements, system design, UI prototype, and anticipated benefits of implementing FairFlow as a scalable solution for community food redistribution.

Functional Requirements

1) ➧ **Data Collection and Processing ~**

   ➧ The system must import and organize data from donation records, distribution logs, and partner agencies.

2) ➧ **Demand Forecasting ~**

   ➧ The system must generate weekly or monthly predictions of food demand by region or partner site.

3) ➧ **Allocation Recommendations ~**

   ➧ The system must recommend how available food should be distributed to different locations based on predicted need.

4) ➧ **Dashboard and Visualizations ~**

   ➧ The system must provide charts, maps, and summaries that allow users to view demand, supply, and distribution activity.

5) ➧ **User Accounts and Permissions ~**

   ➧ The system must allow different user roles with appropriate access levels (enrolled individuals, admin, pantry staff , partner organization, etc).

6) ➧ **Reporting Tools ~**

   ➧ The system must generate downloadable reports summarizing demand, allocations, and key metrics.

7) ➧ **Notifications ~**

   ➧ The system must alert users to shortages, upcoming distribution needs, or data issues.

Non-Functional Requirements

1) ➧ **Performance ~**

➧ The system should load dashboards within a reasonable time

➧ It should handle moderate to large datasets without crashing.

2) ➧ **Security ~**

➧ User accounts must be protected through logins and role based access codes.

➧ Data must be stored securely and transmitted over encrypted connections.

3) ➧ **Usability ~**

➧ The interface should be simple enough for non-technical staff to use.

➧ Visuals and dashboards should be easy to read and interpret.

4) ➧ **Reliability ~**

➧ The system should be available with minimal downtime.

➧ Data should be backed up regularly to avoid loss.

5) ➧ **Scalability ~**

➧ The system should support more users, additional locations, or larger datasets over time.

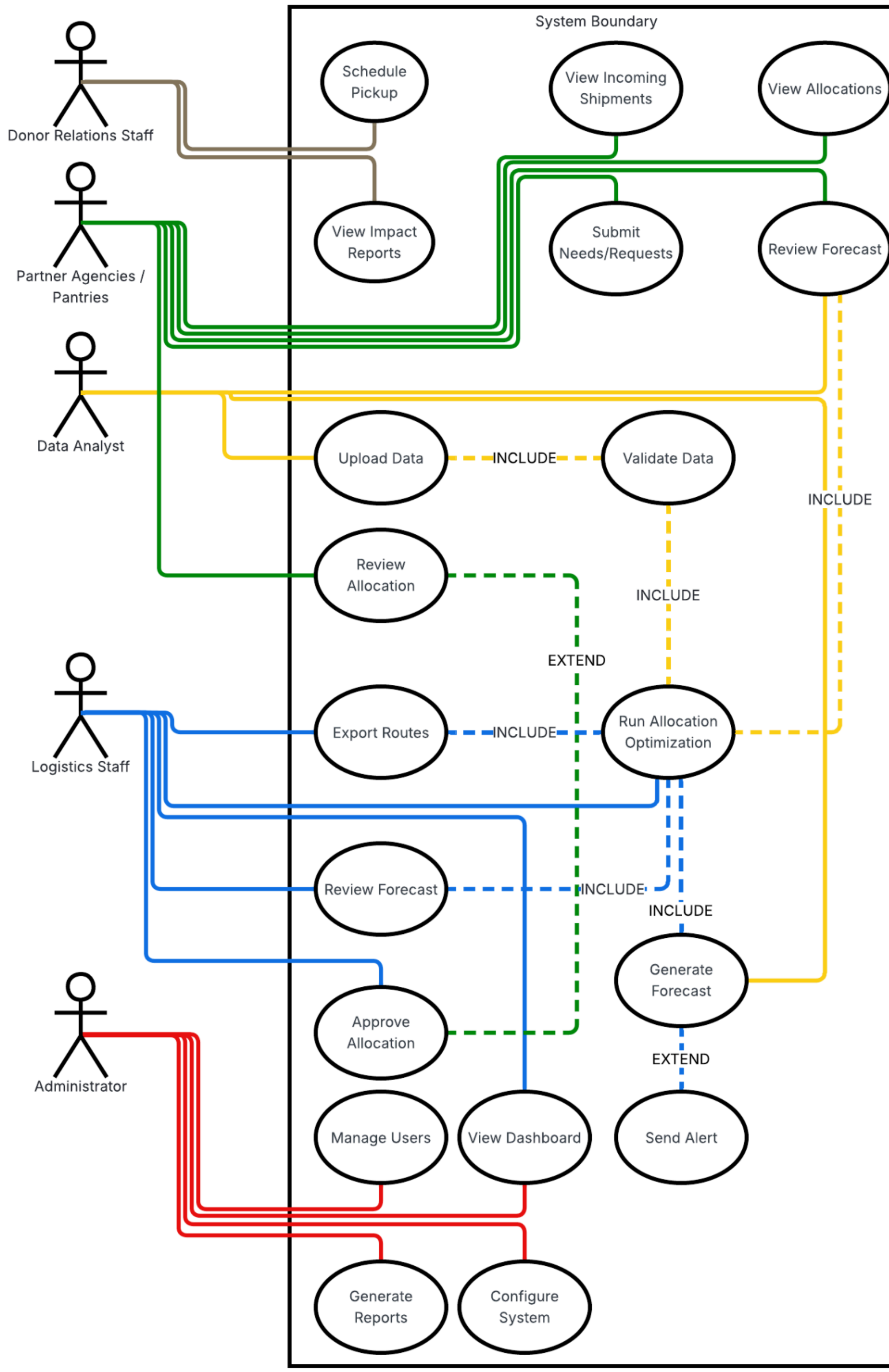
6) ➧ **Maintainability ~**

➧ The system should be modular so updates and fixes can be done easily.

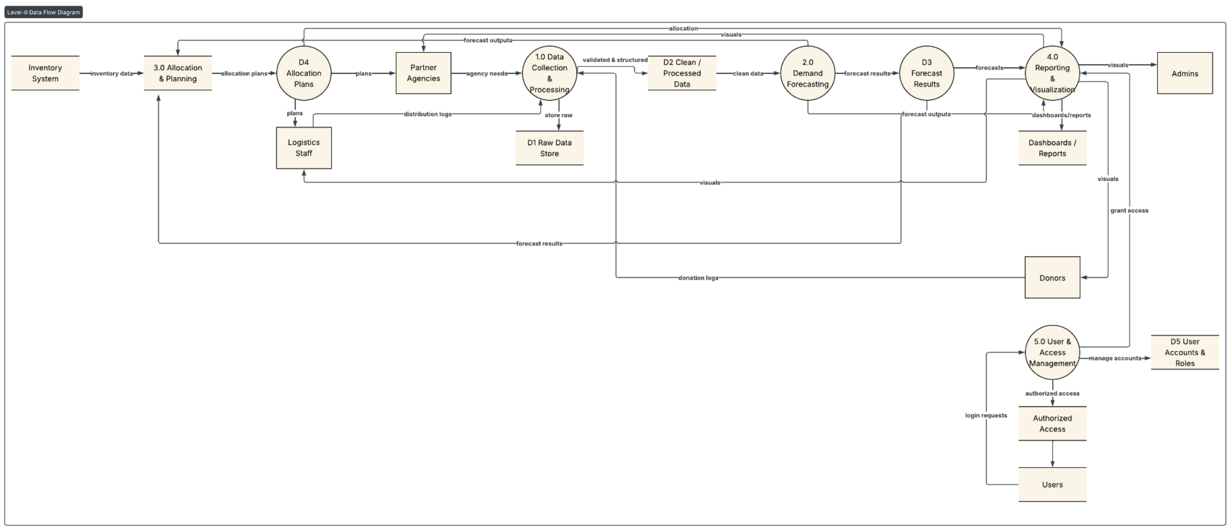
Stakeholder Register

Stakeholder Role	Category	Interest	Expectation
Food Bank Administrators	Internal	Strong	➧ Need clear visibility into demand, supply, and overall performance. ➧ Need reports to support planning and decision-making.
Logistics / Pantry Operations Staff	Internal	Neutral	➧ Need accurate demand forecasts and allocation suggestions. ➧ Need tools that help plan deliveries and manage inventory.
Partner Agencies / Pantries	External	Strong	➧ Need visibility into what food they will receive and when. ➧ Need a simple way to communicate needs or shortages.
Data Analysts	Internal	Strong	➧ Need access to reliable, well-structured data for analysis. ➧ Need consistent outputs from forecasting and reporting tools.
Donor Relations Staff	External	Strong	➧ Need summary reports showing impact and distribution outcomes.
IT / System Administrators	Internal	Neutral	➧ Need tools to manage users, maintain uptime, and ensure security.

Use Case Diagram



Level-0 Data Flow Diagram



Processes ~

1) ➧ **Collect and Validate Data ~**

➧ Gathers pantry requests, inventory updates, and donor information, then checks for completeness and accuracy.

2) ➧ **Generate Forecasts ~**

➧ Uses validated data and historical trends to estimate future food demand.

3) ➧ **Optimize Allocation ~**

➧ Determines how food should be distributed based on forecasts, inventory, and constraints.

4) ➧ **Manage Routing and Distribution ~**

➧ Converts allocation plans into delivery routes, sends assignments to drivers, and records delivery confirmations.

5) ➧ **Reporting and Dashboards ~**

➧ Compiles system data into summaries and visual reports for administrators, analysts, donors, and partner agencies.

Data Stores ~

1) ➧ **Inventory Data ~**

➧ Current stock, donations, and incoming supplies.

2) ➧ **Pantry Requests ~**

➧ Submitted needs and historical demand records.

3) ➧ **Forecast Data ~**

➧ Stored forecasting inputs and past projections.

4) ➧ **Allocation and Routing Data ~**

➧ Allocation plans, schedules, routes, and delivery updates.

5) ➧ **User and System Data ~**

➧ User accounts, roles, and configuration settings.

Data Flows ~

1) ➧ **Incoming Data ~**

➧ Pantry needs, donation updates, external data, and user inputs.

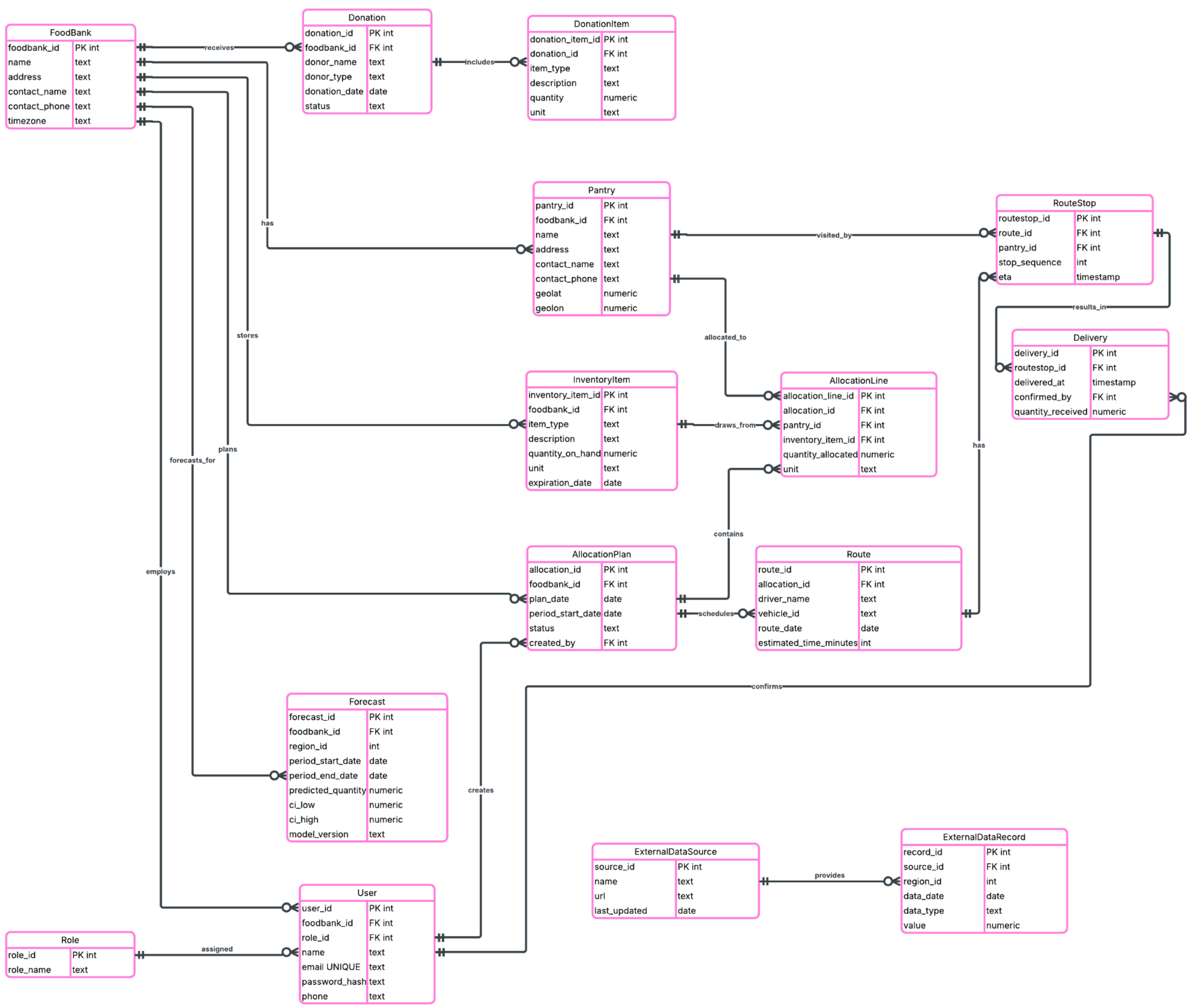
2) ➧ **Internal Flows ~**

➧ Validated data → forecasts → allocation plans → routing → delivery updates.

3) ➧ **Outgoing Data ~**

➧ Status updates to pantries, dashboards to staff, and alerts for shortages or delivery issues.

ERD using Crowsfoot Notation



System Architecture Design

FairFlow uses a cloud based three tier architecture consisting of a presentation layer, an application layer, and a data layer. This architecture was chosen because it is easy to scale, supports multiple users, and allows the system to handle forecasting, reporting, and data processing without slowing down. It also separates responsibilities, making the system easier to maintain and update over time.

<u>Proposed System Architecture ~</u>

1) **Presentation Layer (Client Side) ~**

   ➧ Web browser and app interface used by administrators, logistics staff, and partner pantries.

   ➧ Displays dashboards, maps, reports, inventory, deliveries and allocation recommendations.

2) **Application Layer (Server Side) ~**

   ➧ Handles business logic, demand forecasting, and allocation calculations.

   ➧ Provides APIs for the interface to request data and submit updates.

3) **Data Layer (Storage) ~**

   ➧ Central database that stores inventory, pantry requests, forecasts, deliveries and reports.

   ➧ File storage for uploaded data, historical reports, and model outputs.

   ➧ This structure supports real time updates, clean data flow, and future expansion if FairFlow grows.

Hardware & Software Specifications

1) Software ~

- Frontend: App (JavaScript) Web application (HTML, CSS, JavaScript).
- Backend: Server side application (Python).
- Database: MySQL.
- Cloud Hosting: AWS or Azure

2) Hardware ~

- Cloud server (application layer):
    - 2–4 vCPUs, 8–16 GB RAM.
- Database server:
    - Managed cloud database, 200–300 GB storage.
- File storage:
    - Cloud storage bucket for reports and data uploads.

Conclusion and Recommendations

Conclusion ~

FairFlow is a practical, high impact solution that combines demand forecasting, allocation optimization, and easy to use dashboards to help food banks and partner pantries distribute food more equitably and efficiently. By aligning donations with neighborhood level needs and automating routine logistics decisions, FairFlow can reduce waste, improve coverage in high need areas, and provide transparent reporting for operations and donors.

Key Benefits ~

1) ➧ **Improved equity** ~

   More consistent, data driven allocations reduce geographic and demographic disparities in food access.

2) ➧ **Reduced waste** ~

   Smarter matching of perishables to demand lowers spoilage and disposal costs.

3) ➧ **Operational efficiency** ~

   Automated allocation and routing reduce manual planning time and delivery costs.

4) ➧ **Better decision support** ~

   Dashboards and reports give leaders timely insight for planning and fundraising.

5) ➧ **Stronger donor relationships** ~

   Impact reporting and predictable pickup schedules make large donations easier to absorb.

6) ➧ **Scalable foundation** ~

   Modular design supports growth from a single food bank to regional networks.

Potential Implementation Challenges ~

1) ➧ **Data quality and availability** ~

   Incomplete, inconsistent, or delayed donation and pantry data will set limits to the model accuracy.

2) ➧ **Change management** ~

   Staff and partners may resist new workflows or distrust automated recommendations.

3) ➧ **Privacy and data sharing** ~

   Handling recipient and donor information requires careful aggregation, anonymization, and agreements.

4) ➧ **Technical integration** ~

   Connecting legacy inventory systems, routing tools, or partner platforms can be complex.

5) ➧ **Resource constraints** ~

   Small food banks may lack IT staff, hardware, or budget for initial deployment and ongoing maintenance.

6) ➧ **Model limitations and fairness risks** ~

   Forecasting models can possibly reflect biases in historical data without safeguards, meaning allocations could unintentionally disadvantage some communities.

Conclusion and Recommendations

Recommendations & Next Steps ~

➧ Run a short test (6–12 weeks) with one food bank and a small set of partner pantries.

➧ Validate data pipelines, measure forecast accuracy, test allocation logic, collect user feedback.

➧ Begin with aggregated data from the census level to reduce privacy risk and speed onboarding.

➧ Use aggregated outputs for initial dashboards and donor reports.

➧ Prioritize data quality and consistency before advanced modeling.

➧ Implement basic schema checks, required fields, and simple imputations so forecasts are not undermined by bad data.

➧ Hold frequent training sessions.

➧ Add fairness constraints and transparency features to the allocation engine.

➧ Publish allocation rules, and include per capita and equity metrics.

➧ Plan for lightweight integration rather than heavy system rewrites.

➧ Provide simple import and export paths so partners can adopt the system without changing their core tools.

➧ Track forecasts, delivery, spoilage rates, and equity metrics, then use these to refine models and operations.

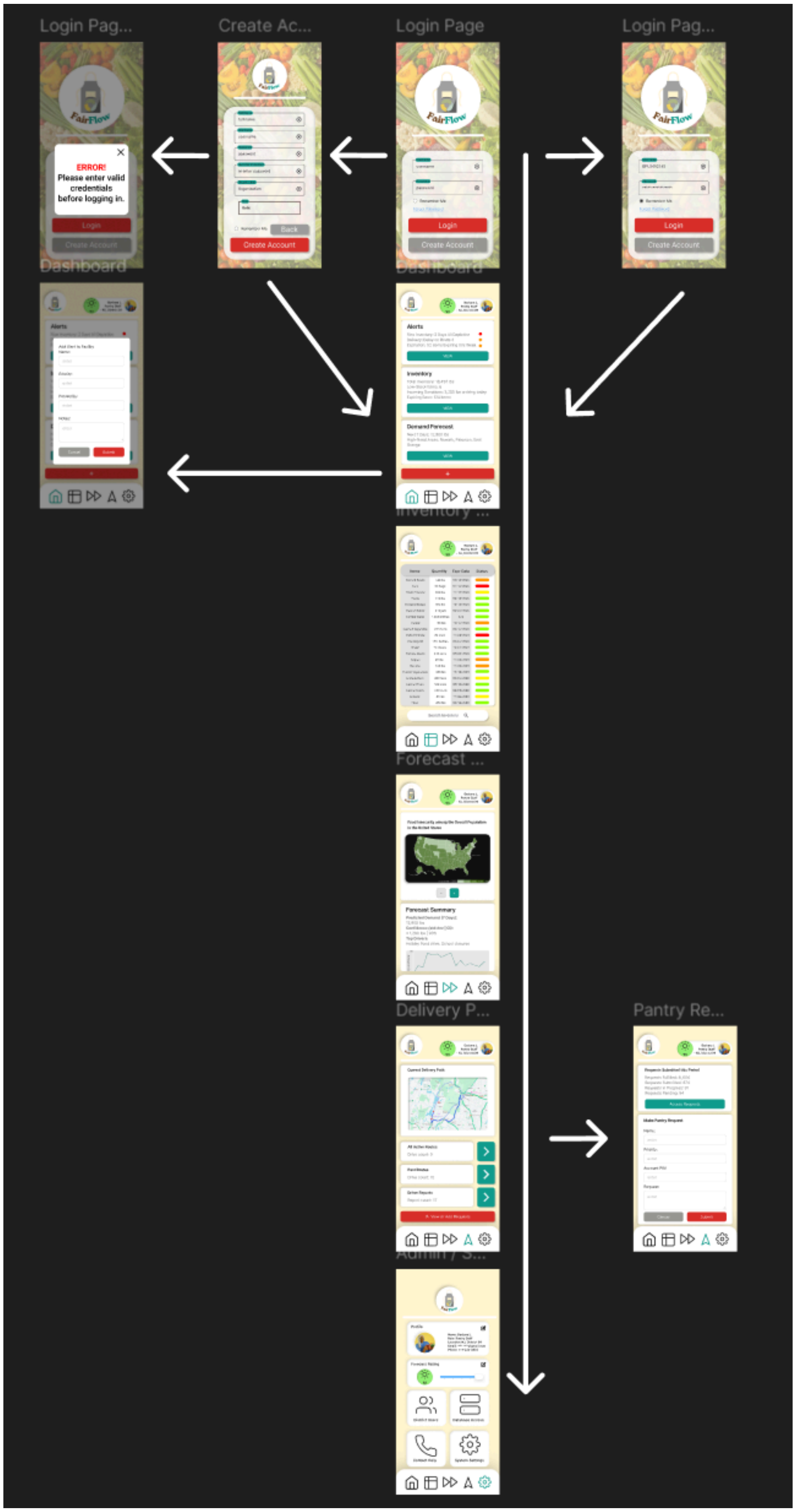
➧ Define success thresholds early.

➧ Secure funding for operations and support to cover hosting, maintenance, and user support.

UI Prototype

[FairFlow Figma Prototype Link]

[Quick YouTube Video Demo Link]



References

Feeding America. (n.d.). *Map the Meal Gap*. Retrieved from https://map.feedingamerica.org Feeding America

Feeding America. (2024, January). *2023 Annual Report*. Retrieved from https://www.feedingamerica.org/sites/default/files/2024-01/FA_23AR_EV_with%20Board_F.pdf Feeding America

Second Harvest Heartland. (n.d.). *Using Data to Drive the Hunger Fight*. Retrieved from https://www.2harvest.org/end-hunger-together/stories/using-data-drive-hunger-fight